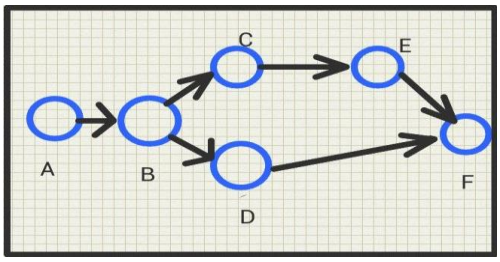


Project Time Management

Topic # 5

Chapter 6 - PMBOK

Chapter 6 - Schwalbe



PERT (Program Evaluation & Review Technique)

- A network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates.
- Probabilistic time periods are used.
- Also referred to as Three point estimation technique.

$$\text{PERT weighted average} = \frac{\text{optimistic time} + 4 \times \text{most likely time} + \text{pessimistic time}}{6}$$

Activity Duration Estimating

- Draw parallels to similar activities
- Historical data
- •Expert advice
- •Delphi technique
- •Three point technique
  - E = O+4M+P/6
- •Rule of Thumb
  - E.g. A task can be completed in 4 days at an optimistic side, 7 days if pessimistic and it requires most likely six days to be completed. What is the estimated duration of the task?

PERT & NETWORK

- a- optimistic, m- most likely, b-pessimistic
- Variance $\sigma^2 = ((b-a)/6)^2$
- Standard deviation: $\sigma = \sqrt{\sigma^2}$
- Uncertainty of project completion time

$$Z = (D - \mu)/\sqrt{\sigma_e^2}$$

and

D = the desired project completion time

μ = the critical time of the project, the sum of the TEs for activities on the critical path

σ_e^2 = the variance of the critical path, the sum of the variances of activities on the critical path

Z = the number of standard deviations of a normal distribution (the *standard normal deviate*)

Example

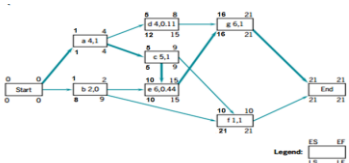
Consider the following project (times given in days).

Activity	*a*	*m*	*b*	*Predecessors*
a	1	4	7	—
b	2	2	2	—
c	2	5	8	a
d	3	4	5	a
e	4	6	8	c,b
f	0	0	6	c,b
g	3	6	9	d,e

Find:

1. The network.
2. All expected activity times, variances, and slacks.
3. The critical path and expected completion time.
4. The probability the project will be done in 23 days.
5. The completion time corresponding to 95% probability.

Solution



2. Activity	TE	σ^2	Slack
a	4	1.00	0
b	2	0	7
c	5	1.00	0
d	4	0.11	7
e	6	0.44	0
f	1	1	11
g	6	1	0

3. Critical path is a-c-e-g for a time of 21 days.
4. $z = (23 - 21)/\sqrt{3.44} = 1.078$ for a probability of 85.9%.
5. P = 0.95 corresponds to $z = 1.65 = (T - 21)/1.855$, or T = 24.06 days.

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

Working days v/s calendar days

- When first planning a project, it is convenient to do all calculations in working days (ignoring holidays and weekends).
- Project Management software will do these calculations and take account of holidays, weekends, etc.
- If an activity starts on August 01, and take 12 days the activity A will finish on August 18:

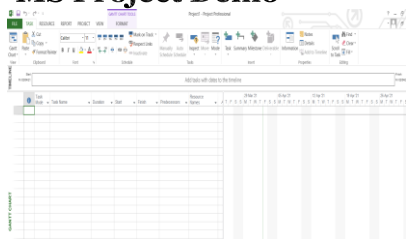
August, 2014

Su	Mo	Tu	We	Th	Fr	Sa
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

Implementing Project Constraints

- Working Days versus Calendar days
- A *constraint* is a boundary or limit based on the project.
- Four types of constraints that can be imposed for creation of network diagram besides natural constraint:
  - date,
  - managerial,
  - technical,
  - organizational constraint.

MS Project Demo



END OF TOPIC 5

COMING UP!!!!!!

Project Cost Management

HR - Project Team Organization

Project Control

12